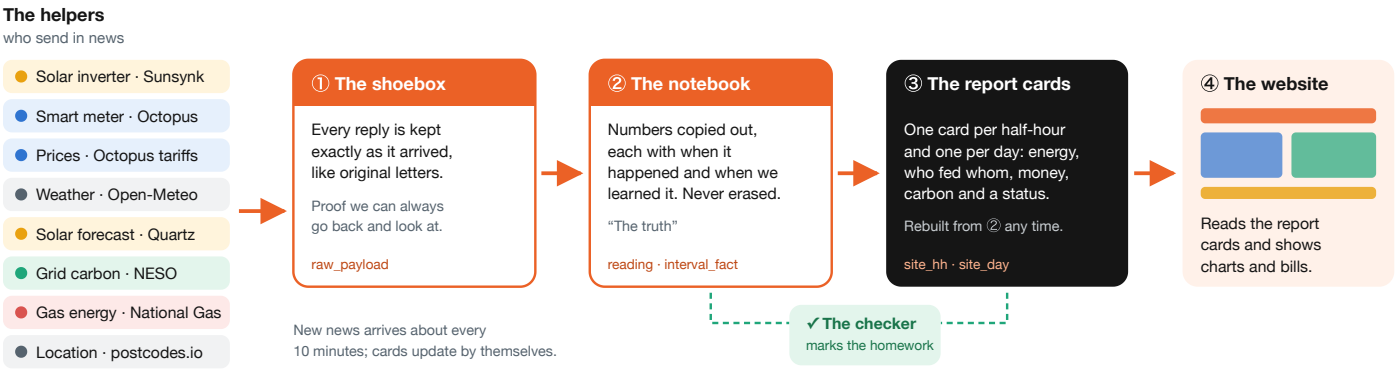
ENERGIEBEE CORE · DATABASE

How the EnergieBee database works

What it remembers, where the information comes from, how the pieces fit together, and how the website uses it. Explained in plain words, no technical background needed.

> **The big idea:** think of the database as a **home's energy diary**. Every few minutes, helpers outside the house (the solar inverter, the smart meter, the energy supplier, the weather service) send in news. The diary keeps every letter it receives, copies the important numbers into a neat notebook, and then writes a **report card for every half-hour**: how much sunshine power was made, how much was bought and sold, what it cost and what it saved. The website reads those report cards.

1 From the outside world to your screen



2 Where the information comes from

Helper	What it tells us	A simple example
Solar inverter (Sunsynk)	Sunshine power made, battery in and out, battery charge level, every 5 minutes.	"At 12:05 the panels made 1.2 kW and the battery was 82% full."
Smart meter (Octopus)	Electricity bought and sold, and gas used, per half-hour. Arrives about a day late.	"Between 18:00 and 18:30 the house bought 0.9 kWh."
Tariff prices (Octopus)	The price of each unit, the daily standing charge and the export price, with dates.	"Night power costs 7.6p; daytime costs 32.1p."
Forecasts (Quartz, Open-Meteo)	How much sun to expect, and the weather and temperature outside.	"Tomorrow should bring about 4 kWh of solar."
Grid carbon (NESO)	How clean the grid's electricity is in the home's region.	"Right now each kWh from the grid made 14 g of CO_2."
Gas energy (National Gas)	How much energy is in a unit of gas, so meter readings can turn into kWh.	"One cubic metre of gas held about 11 kWh today."
Location & catalogue	postcodes.io finds the home's region; Tariffscanner lists tariffs on sale to compare.	"This postcode is in grid region G."

Each helper has a **"connector"** that knows how to talk to it. The home never calls a helper directly; the database records which helper fills which job for each country.

3 The six drawers: what the database stores

The database has **53 tables** (a table is a labelled list, like a spreadsheet tab), kept in six drawers:

🏠 WHO & WHERE

The home, its country, region, currency and time zone; its meters, inverter and car; and which helpers it is connected to.

`site · device · connection · market · region`

💷 PRICES

Every tariff, its prices over time, taxes such as VAT, and which tariff each meter was on, on which dates.

`tariff · rate_interval · tariff_agreement · tax_rule`

📒 THE TRUTH

The shoebox of original replies, plus every number taken from them: readings, half-hour amounts and forecasts.

`raw_payload · reading · interval_fact · forecast_fact · series`

📊 REPORT CARDS

The ready-to-show results: one row per half-hour and one per day, with energy, money, carbon and status.

`site_hh · site_day`

✅ CHECKS & HISTORY

Each day's cross-checks, every time a day's figures changed and why, corrections, and a log of each sync.

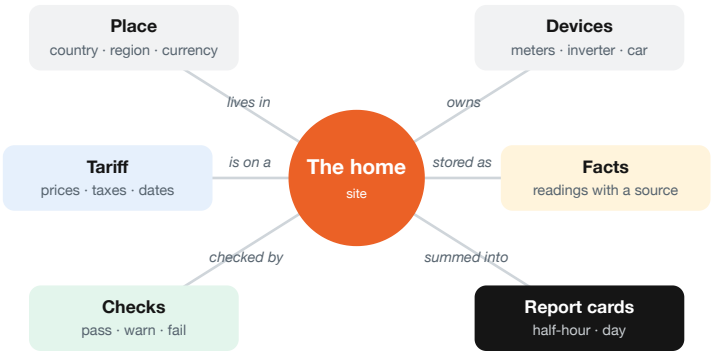
`check_result · site_day_revision · fact_correction · sync_run`

🔮 READY FOR LATER

Space for future battery plans and device commands. The shelves are built; the planner that fills them is not built yet.

`plan_run · plan_slot · command`

4 How the tables hold hands



Everything hangs off the home. A home lives in a place, owns devices and is on a tariff. Each device's numbers are stored as facts, and every fact keeps a label saying which helper sent it.

Follow one number: the smart meter says “0.9 kWh bought at 18:00”. That fact joins the tariff price for 18:00 (32.1p) to make the report card's cost: **about 29p**. The checker then compares the meter with the inverter's own clamp to make sure they agree.

5 Three golden rules

✏️ Never rub out

If a helper sends a correction, it is added as a new version; the old one is kept. We can always say “what did we know on Tuesday?”

🏷️ Always say where from

Every number points back to the original reply it came from, and unchecked or demo values are clearly labelled.

🚫 Never make things up

Missing data stays missing. A half-hour is **final**, **provisional** (meter data still on its way) or **missing**.

6 How the website uses it

The website **only reads the report cards and facts**; it never asks the helpers. The bill on *My energy* is the cards added up; *Solar*, *Battery* and *Grid* draw them as charts; *Data & trust* and *Data Explorer* open the notebook and shoebox so anyone can check a number.

When late meter data or a new price arrives, the affected days go on a **to-do list** (`rebuild_queue`) and their cards are rebuilt from the notebook. Same facts, same answer; every change to a finished day is written into its history.

7 The diary today · live prototype, 26 Oct 2026: one home in Great Britain, data since 9 Oct

53	3,264	80,000+	837	6
tables	original replies kept	facts in the notebook	half-hour report cards	countries ready: GB FR NL DE BE ES